Student number_____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

The First Semester of 2021-2022 Year

Probability Theory and Mathematical Statistics Middle exam

Seat No. _____

(Closed book Time 120 minutes)

Question number	I	II	III	IV	Total score
Score					
Marker					

Score	

I. Completions (5 items, 2 points each)

1. Events C_1, C_2, and C_3 are mutually independent, and $P(C_1)= P(C_2)=P(C_3)=1/3$, then the value of $P[(C_1 \cap C_2) \cup C_3]$ is_____.

2. Suppose that the probability of encountering a red light at a crossroad is 0.2, and traffic lights at each crossroad are independent of each other. What is the probability of a man encounters a red light for the second time when he walks through the fifth crossroad?_____.

3. Suppose that the pdf of r.v. $\boldsymbol{X}$ is $\boldsymbol{f}(\boldsymbol{x}) = \begin{cases} 2x & 0 \le x < 1 \\ 0 & \text{others} \end{cases}$, then the probability of $\boldsymbol{X}$ in interval (0.3,0.7) is_____.

4. Suppose that the number of calls（$\boldsymbol{X}$）received by the service center per minute has a Poisson distribution with a parameter of λ=3, Then the probability of receiving more than 3 calls in one minute is_____.

5. Suppose that the random variable $\boldsymbol{X}$ has an exponential distribution with a parameter of 2. If we conduct four independent observations of $\boldsymbol{X}$, then the probability that three observations of $\boldsymbol{X}$ are larger than 5 is_____.

II. Choices (5 items, 2 points each)

Score	

6. Suppose that $\boldsymbol{A}$, $\boldsymbol{B}$ are two random events, $P(\boldsymbol{AB})=0$, then the correct conclusion is ( ).
   A. Event A and B are exclusive B. AB is an impossible event
   C. AB is a possible event D. P(A)=0 or P(B)=0

7. Given that $P(\boldsymbol{A}) = 0.6$, $\mathrm{P}(\boldsymbol{AB}) = 0.2$, and $\mathrm{P}(\boldsymbol{B}) = 0.3$, then $\mathrm{P}(\mathbf{A}|\mathbf{B})$ is ( ).
   A. $1/4$ B. $3/5$
   C. $1/3$ D. $2/7$

8. Suppose components $\boldsymbol{A_1}$, $\boldsymbol{A_2}$, $\boldsymbol{B_1}$ and $\boldsymbol{B_2}$ operate independently in the system shown in Figure 1. Assume that all the components in the system start operating at the same time and a component does not work again once it fails. The probability of functioning for each of the components is $\boldsymbol{p}$. The entire system works (i.e.,operates without failure) if $\boldsymbol{A_1}$ or $\boldsymbol{A_2}$ works and $\boldsymbol{B_1}$ or $\boldsymbol{B_2}$ works. Then the probability that the entire system works is ( ).

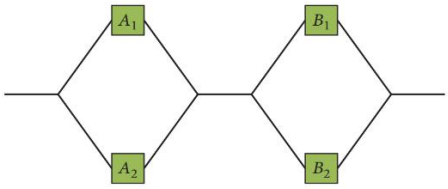

Figure 1. System components.

   A. $\boldsymbol{p}^4$ B. $1-(1-\boldsymbol{p})^4$
   C. $(1-\boldsymbol{p}^2)(1-\boldsymbol{p}^2)$ D. $(2\mathrm{p}-\boldsymbol{p}^2)^2$

9. If $\boldsymbol{f}(\boldsymbol{x})$ and $\boldsymbol{F}(\boldsymbol{x})$ are the probability density function and cumulative distribution function of random variable $\boldsymbol{X}$, which of the following option is correct? ( )
   A. $P(\boldsymbol{X} = \boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x})$ B. $P(\boldsymbol{X} = \boldsymbol{x}) \leq \boldsymbol{F}(\boldsymbol{x})$
   C. $0 \leq \boldsymbol{f}(\boldsymbol{x}) \leq 1$ D. $P(\boldsymbol{X} = \boldsymbol{x}) = \boldsymbol{F}(\boldsymbol{x})$

10. If the cumulative distribution function of random variable $\boldsymbol{X}$ is

$$\boldsymbol{F}(x) = \begin{cases} A + B\frac{1}{x}, \boldsymbol{x} > 1 \\ 0,\ x \leq 1 \end{cases}, \text{ then ( )}$$

   (A) $A = 1,\ B = -1$ (B) $A = -1,\ B = 1$
   (C) $A = 2,\ B = -2$ (D) $A = -2,\ B = 1$

III. Calculations (6 items, 12 points each)

Score	

11. A box contain four fair dice. Two have faces numbered 1, 2, 3, 4, 5, 6; one has faces numbered 2, 2, 4, 4, 6, 6; and one has all six faces numbered 6. Now taking one dice randomly in this box and rolling it two times independently.

Calculate:

(1) The probability that two 6 are rolled;

(2) If two 6 are rolled, find the probability that the dice is the one that has all six faces numbered 6.

12. Continuous random variable X has a probability density function of

$$f(\mathrm{x}) = \begin{cases} Acos(x), & |x| \le \frac{\pi}{4}, \\ 0, & \text{otherwise} \end{cases}.$$

Calculate:

(1) The value of A and the cumulative distribution function of r.v. X;

(2) The probability that equation $t^2 - \frac{\mathrm{t}}{\sqrt{3}} + \frac{\boldsymbol{X}}{2\pi} = 0$ has real root.

13. The probability distribution of r.v.$\boldsymbol{X}$ is known as:

$\boldsymbol{X}$	0	1	2	3	4	5
p	1/12	1/6	1/3	1/12	$\boldsymbol{k}$	1/9

Find:

(1) The value of $\boldsymbol{k}$;

(2) The probability distribution of $\boldsymbol{Y}=(\boldsymbol{X}-3)^2$.

14. The possible value of $r.v.$ $\boldsymbol{X}$ is 1, 2, 3 and 4, and the four possible values are equally likely. $\boldsymbol{Y}$ is a random variable and its value are integers among $[1, \boldsymbol{X}]$.

Find:

(1) The probability distribution of $(\boldsymbol{X}, \boldsymbol{Y})$;

(2) Two marginal distribution of $(\boldsymbol{X}, \boldsymbol{Y})$.

15. If the joint pdf of 2-D r.v. ($\boldsymbol{X}$, $\boldsymbol{Y}$) is $\boldsymbol{f}(\boldsymbol{x},\boldsymbol{y}) = \begin{cases} Cx^2y, & x^2 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$,

Calculate:

(1) The constant C; (2) P($\boldsymbol{X}$>$\boldsymbol{Y}$);

(3) The two marginal distributions of ($\boldsymbol{X}$, $\boldsymbol{Y}$).

16. Placing a temperature regulator in a container containing a liquid. When the regulator is set as d ℃, the temperature $\boldsymbol{X}$ of the liquid (in ℃) is a random variable, and $\boldsymbol{X}$~$\boldsymbol{N}$(d, 0.4^2). (There are some values you may need: Φ(2) =0.9773, Φ(2.2904)=0.9890, Φ(2.3263)=0.9900, Φ(2.5)=0.9938).

Calculate:

(1) If d = 90, find the probability that $\boldsymbol{X}$ is less than 89;

(2) If the probability of keeping the liquid temperature at least 80 is not less than 0.99, what is the minimum value of d?

IV. Comprehensive analysis Problem (8 points)

Score	

17. The total running time (unit: hour) of an electronic computer before destruction is a continuous random variable, and its probability density function is:

$$f(x) = \begin{cases} \lambda e^{-\frac{x}{100}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Calculate:

(1) The value of λ;

(2) The probability that the computer will run between 50 to 150 hours before destruction;

(3) Picking one of these electronic computers randomly, what is the probability of it can work more than 100 hours.

(4) One of these electronic computers has worked 100 hours, what is the probability that it can work another 100 hours?

Stony Brook Institute at Anhui University
The First Semester of 2021-2022 Year
Probability Theory and Mathematical Statistics
Middle Exam Answers

I. Completions (5 items, 2 points each)

1.11/27. 2. 0.08192 3. 0.4 4. $1 - 13e^{-3}$ 5. 10. $4e^{-30}(1 - e^{-10})$

II. Choices (5 items, 2 points each)

6. one of A,B,C 7. C 8. D 9. B 10. A

III. Calculations and Proofs (6 items, 12 points each)

11. Let event B be two 6 are rolled, A_i be the number i dice has been taken, i=1,2,3,4.

1) $P(B) = \sum_{i=1}^{4} P(B|A_i)P(A_i) = \frac{1}{4} \cdot (\frac{1}{36} + \frac{1}{36} + \frac{1}{9} + 1) = \frac{7}{24}$. (6 points)

2) $P(A_4|B) = \frac{P(A_4B)}{P(B)} = \frac{1/4 \cdot 1}{7/24} = \frac{6}{7}$. (12 points)

12. 1) $1 = \int_{-\infty}^{\infty} f(x)dx = \int_{-\pi/4}^{\pi/4} Acos(x)dx = \sqrt{2}A, \Rightarrow A = \frac{1}{\sqrt{2}}$.

$$F(x) = \int_{-\infty}^{x} f(x)dx = \begin{cases} 0, & x < \frac{\pi}{4}, \\ \int_{-\infty}^{x} \frac{1}{\sqrt{2}} costdt = \frac{1}{\sqrt{2}}(sinx + \frac{1}{\sqrt{2}}) & x \in [-\frac{\pi}{4}, \frac{\pi}{4}] \\ 1, & x > \frac{\pi}{4}. \end{cases}$$ (6 points)

2) the equation has real root" equals to $\triangle = (-\frac{1}{\sqrt{3}})^2 - \frac{4X}{2\pi} \geq 0 \Rightarrow X \leq \frac{\pi}{6}$.

$P(X \leq \frac{\pi}{6}) = F(\frac{\pi}{6}) = \frac{2 + \sqrt{2}}{4}$. (12 points)

13. 1) $1 = \sum p_i = 1/12 + 1/6 + 1/3 + 1/12 + k + 1/9 \Rightarrow k = 2/9$. (6 points)

2)

X	0	1	2	3	4	5
P	1/12	1/6	1/3	1/12	2/9	1/9
$Y = (X-3)^2$	9	4	1	0	1	4

$$\Longrightarrow \quad Y \sim \begin{pmatrix} 0 & 1 & 4 & 9 \\ 1/12 & 5/9 & 5/18 & 1/12 \end{pmatrix} \qquad \text{(12 points)}$$

14. 1)

$X \setminus Y$	1	2	3	4
1	1/4	0	0	0
2	1/8	1/8	0	0
3	1/12	1/12	1/12	0
4	1/16	1/16	1/16	1/16

(6 points)

2) $X \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix} \quad Y \sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 25/48 & 13/48 & 7/48 & 1/16 \end{pmatrix}$ (12 points)

15. 1) $1 = \int_{-\infty}^{\infty} f(x,y)dx = \int_{-1}^{1} cx^2 dx \int_{x^2}^{1} ydy = \dfrac{4c}{21} \Rightarrow c = \dfrac{21}{4}.$ (4 points)

2) $P(X > Y) = \int_0^1 21/4x^2 dx \int_{x^2}^{x} ydy = \dfrac{3}{20}.$ (8 points)

3) The margianl pdf f(x) of r.v.X is

$f(x) = \int_{-\infty}^{\infty} f(x,y)dy = \dfrac{21}{4} \int_{x^2}^{1} x^2 ydy = \dfrac{21}{8}(x^2 - x^6).\ \ 1- \leqslant x \leqslant 1.$

$f(y) = \int_{-\infty}^{\infty} f(x,y)dx = \dfrac{21}{4} \int_{-\sqrt{y}}^{\sqrt{y}} x^2 ydy = \dfrac{7}{2}y^{2.5}.\ \ 0 \leqslant y \leqslant 1$ (12 points)

16. 1) $P(X \leqslant 8.9) = P(\dfrac{X-d}{0.4} \leqslant \dfrac{89-d}{0.4}) = \Phi(\dfrac{89-90}{0.4}) = \Phi(-2.5) = 1-\Phi(2.5) = 0.0062.$ (6 points)

2) $P(X \geqslant 80) = 1 - P(X \leqslant 80) = 1 - P(\dfrac{X-d}{0.4} \leqslant \dfrac{80-d}{0.4}) = 1 - \Phi(\dfrac{80-d}{0.4}) = \Phi(-\dfrac{80-d}{0.4}).$

$\Phi(-\dfrac{80-d}{0.4}) > 0.99, \ \Rightarrow -\dfrac{80-d}{0.4} \geqslant 2.3263 \Rightarrow d = 80.931.$ (12 points)

17.1) $1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^{\infty} \lambda e^{-0.01x} dx \Rightarrow \lambda = 0.01.$ (2 points)

2)$F(x) = \int_{-\infty}^{x} f(t)dt = \int_0^x 0.01e^{-0.01t}dt = 1 - e^{-0.01x}, \Rightarrow F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-0.01x}, & x \geq 0 \end{cases}$

$P(50 \leqslant X \leqslant 150) = F(150) - F(50) = e^{-0.5} - e^{-1.5}.$ (4 points)

3) $P(X > 100) = 1 - P(X \leqslant 100) = 1 - F(100) = 1/e.$ (6 points)

4)$P(X > 200|X > 100) = \dfrac{P(X > 200)}{P(X > 100)} = \dfrac{e^{-2}}{e^{-1}} = e^{-1}.$ (8 points)